REMOTE ACCESS OF FLOATSAT KIT

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ORION Space
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Procedure for remote access of FloatSat

This document contains information to access a computer remotely. We hereby advise you that any unauthorized action might result in civil or criminal liability.

Contact to *orionsapce.nepal@gmail.com* if you came across this document by mistake.

Thank you for your kind cooperation.

PS: Please contact orionsapce.nepal@gmail.com to get password to access the FloatSat kit.

Procedure for remote access of FloatSat

1. Write a code for STM32 on your laptop and generate .hex file.
2. Use *WinSCP* to remotely copy the .hex file from your laptop to Raspberry Pi to which STM32 is connected.
3. Log in to the Raspberry Pi using *PuTTY* and flash the .hex file to STM32.
4. Use any command line based serial viewer like *picocom* to interface with STM32.

SOFTWARES REQUIRED

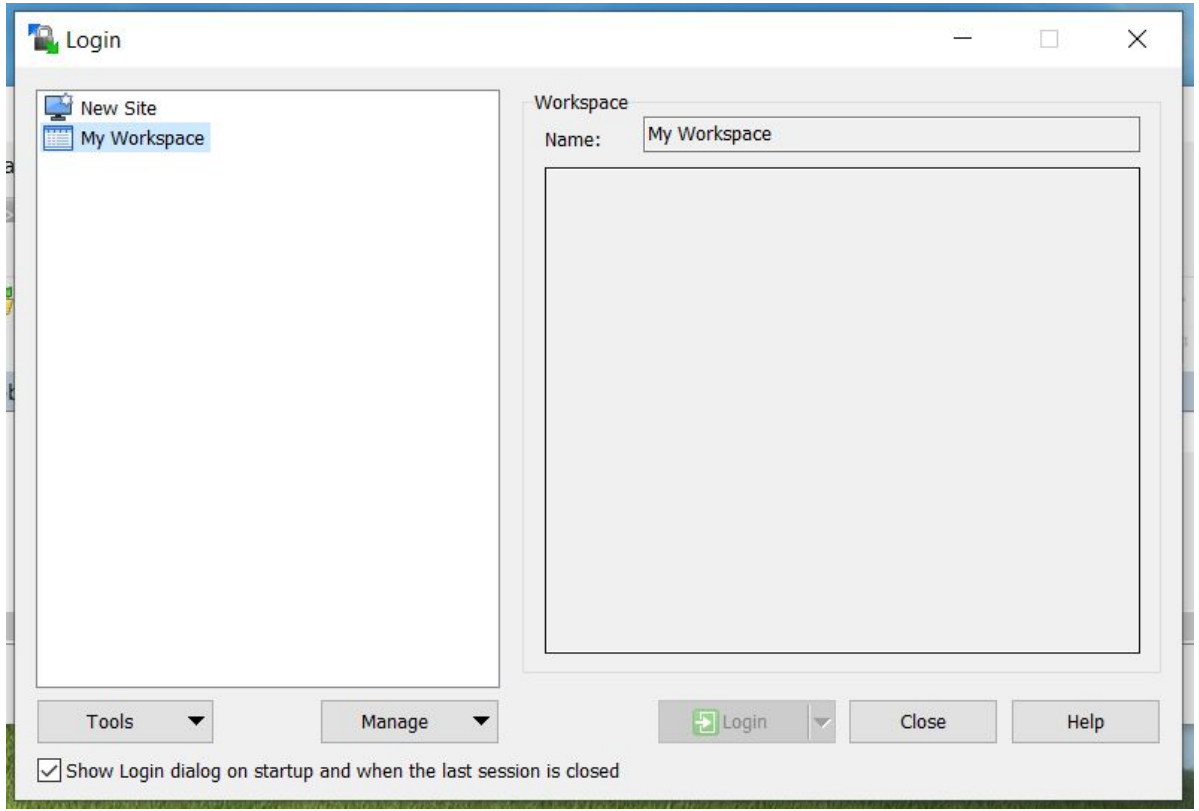
1. PuTTY - a free SSH and telnet client for Windows

   https://www.chiark.greenend.org.uk/~sgtatham/putty/latest.html

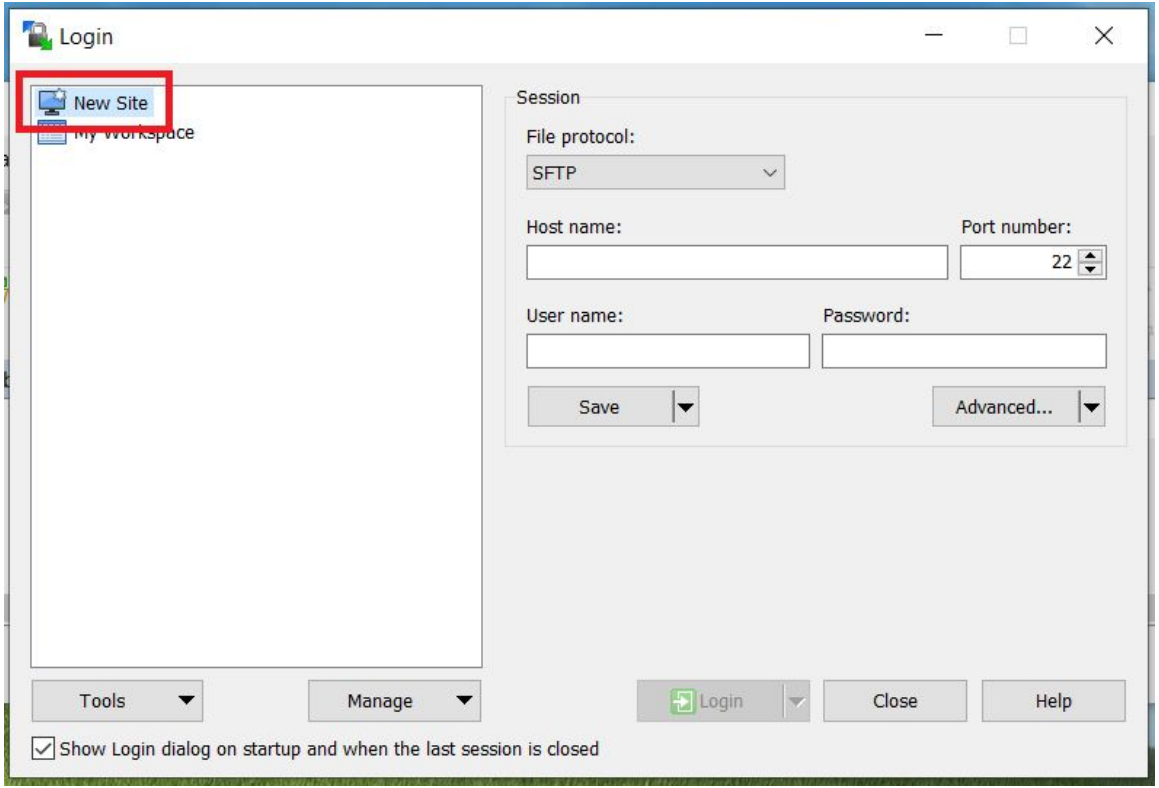
2. WinSCP - WinSCP is a popular SFTP client and FTP client for Microsoft Windows

   https://winscp.net/eng/download.php

Open WinSCP

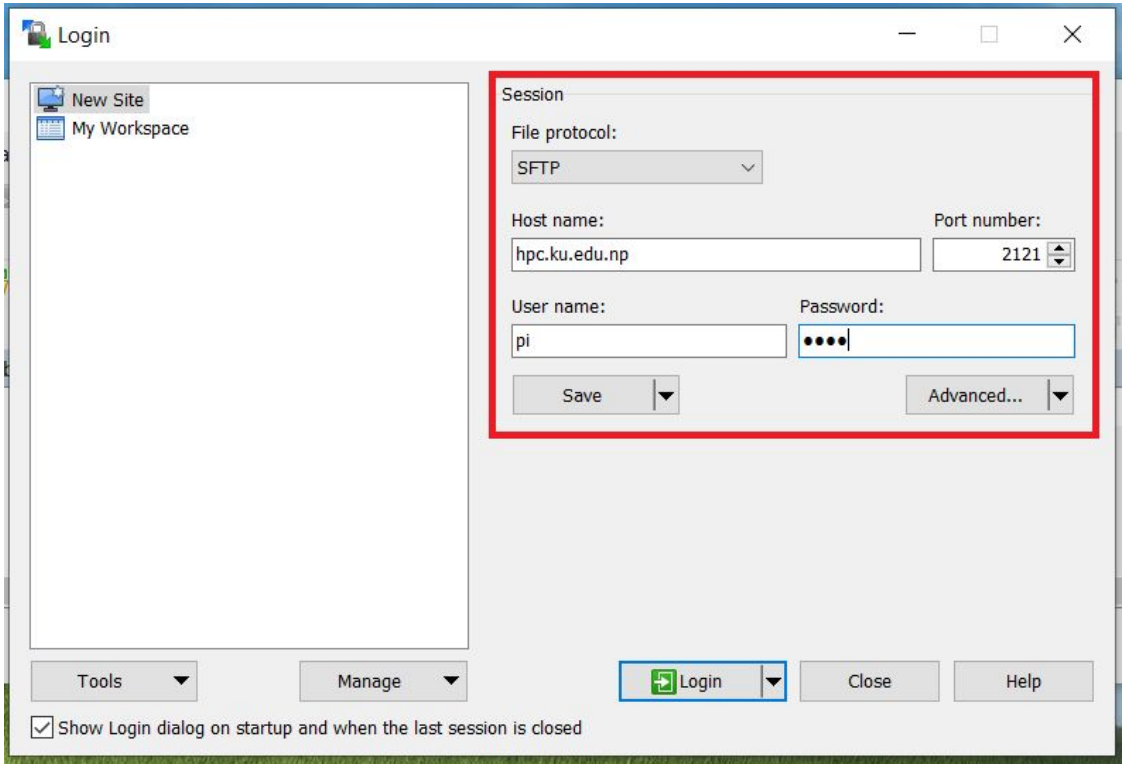

Login
New Site
My Workspace
Workspace
Name:
My Workspace
Tools
Manage
Login
Close
Help
Show Login dialog on startup and when the last session is closed

Go to “New Site”

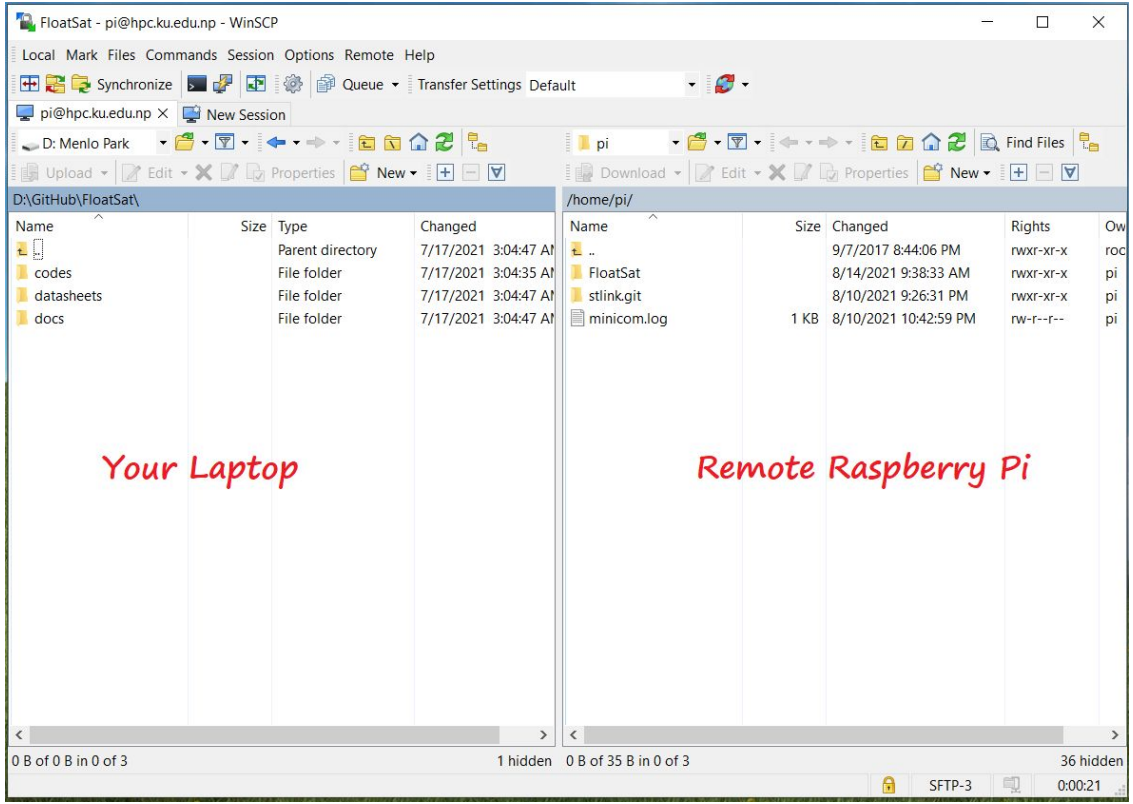


Contact us for password !!

Enter following details



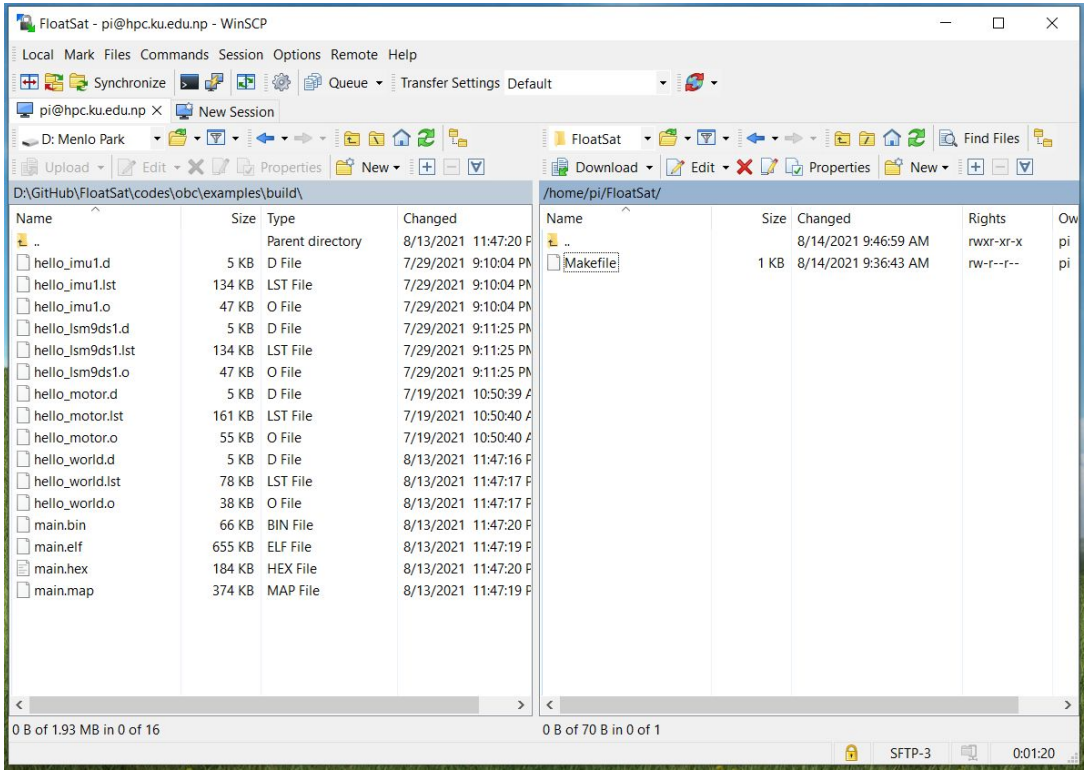
You are logged in to Raspberry Pi



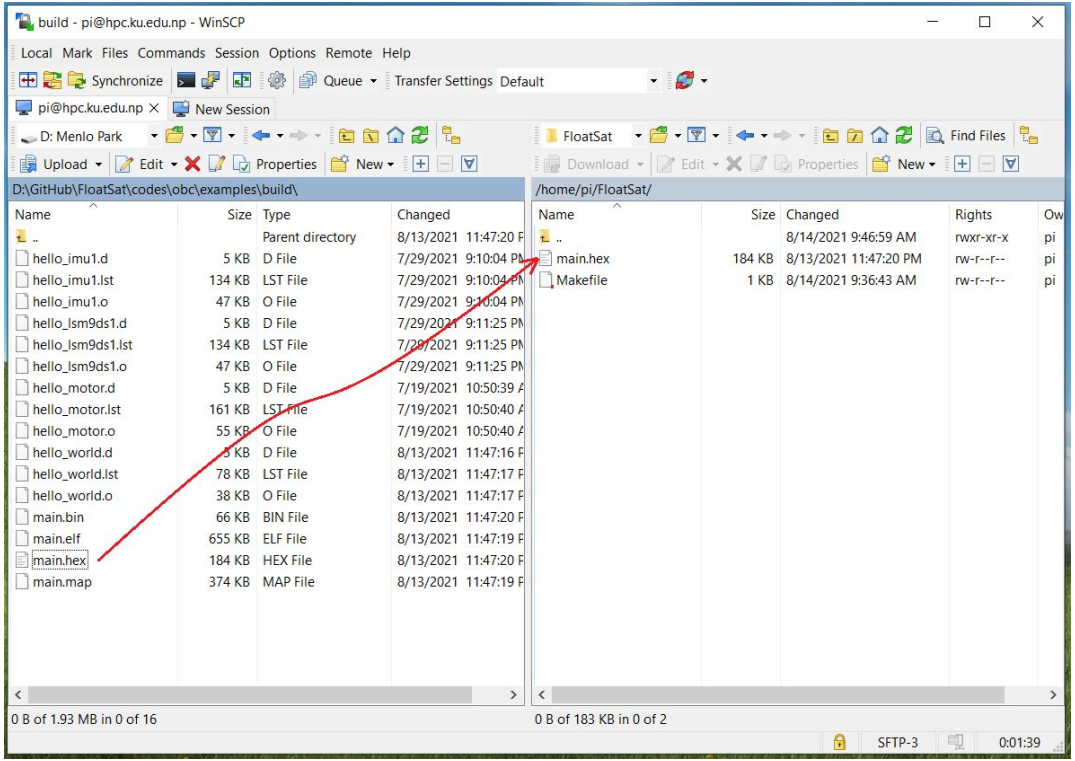
File transfer using WinSCP

After you are logged in to the Raspberry Pi, you can transfer file between your laptop and Raspberry Pi by just drag and drop. In our case, we want to transfer .hex file so open the location of .hex file in your laptop and drag it inside *FloatSat* folder in Raspberry Pi.

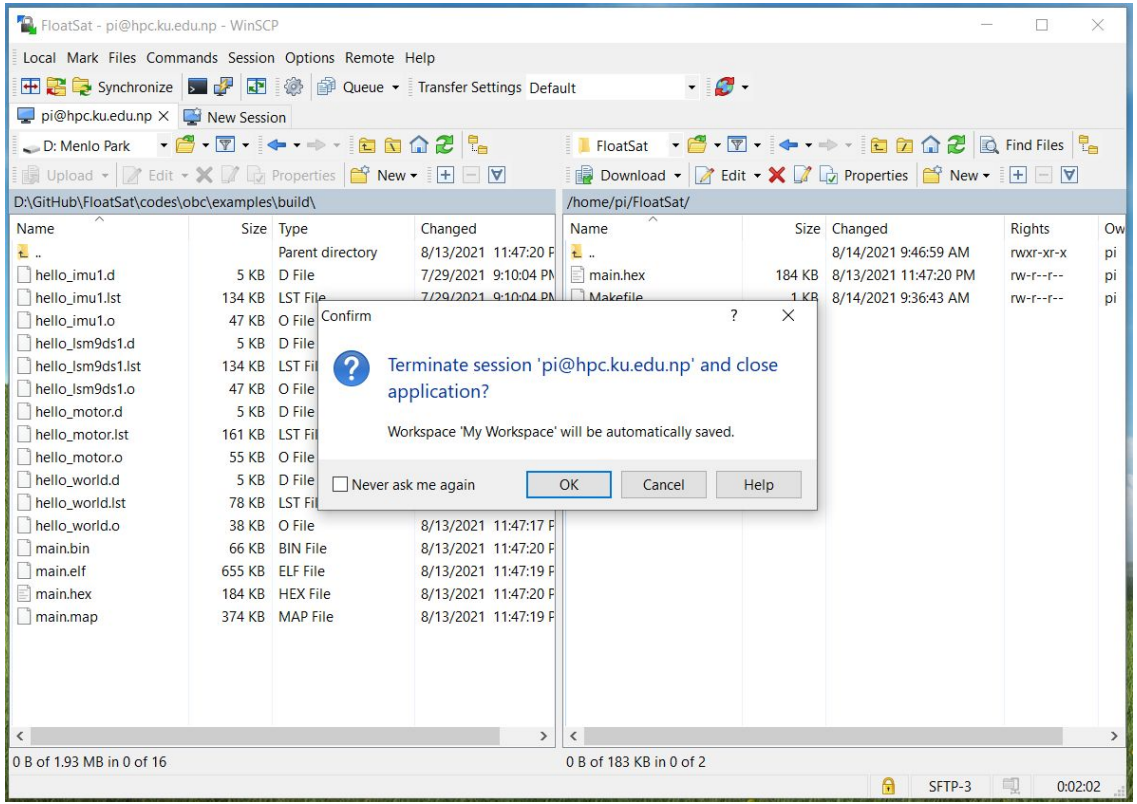
Open *FloatSat* folder in Raspberry Pi



Transfer .hex file by drag and drop



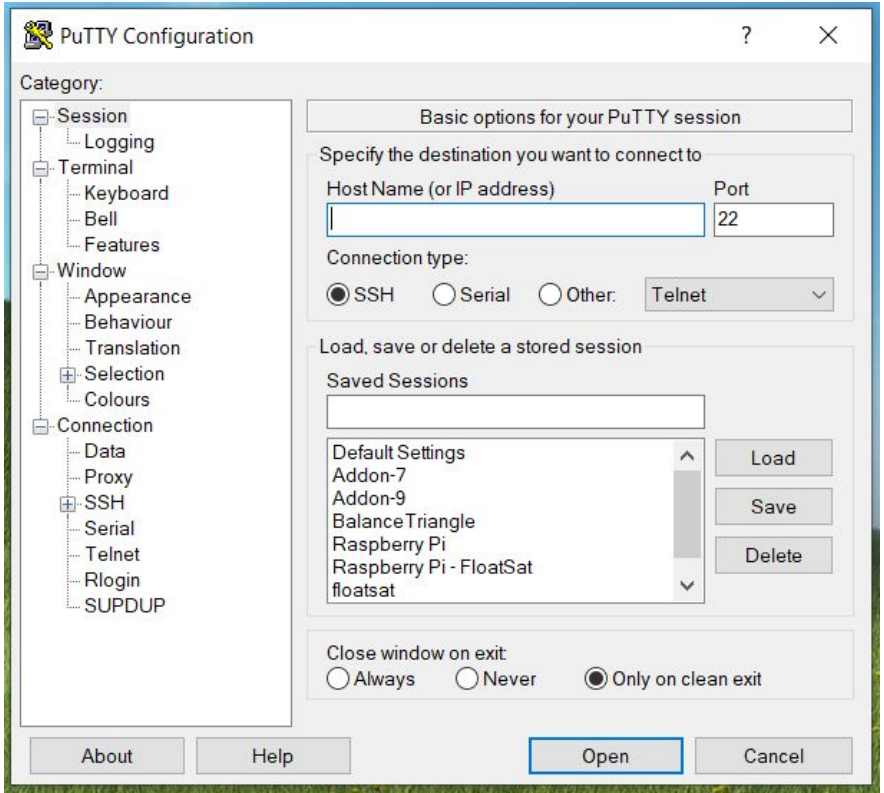
Close WinSCP



Working with PuTTY

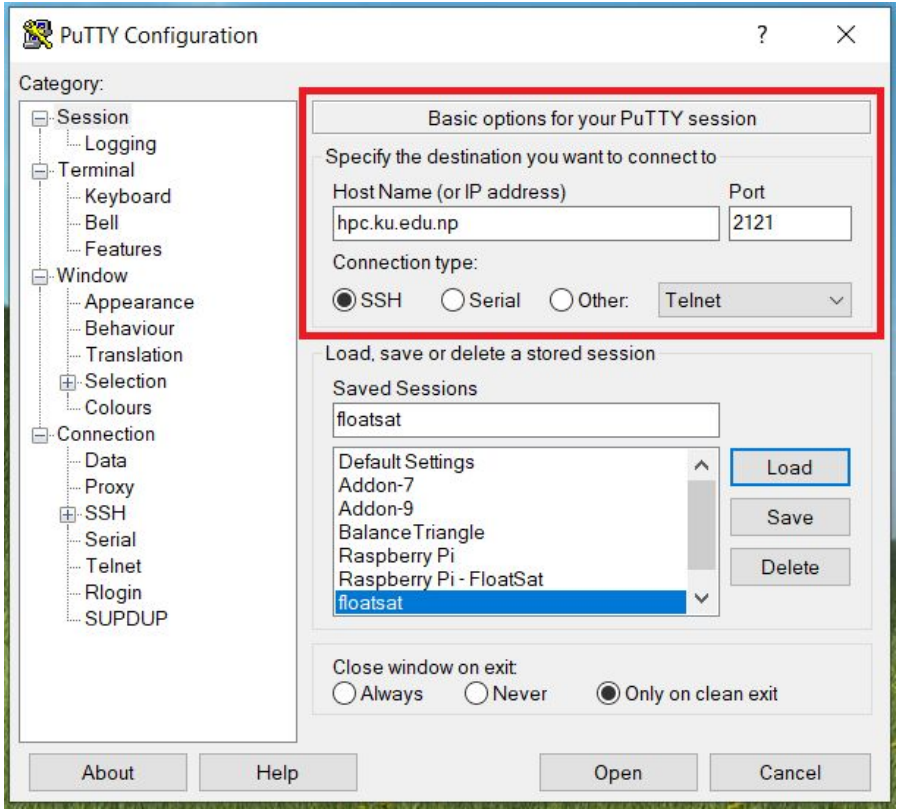
Upto this point you have transferred the .hex file from your laptop to Raspberry Pi. Now all it remains is to upload this file to the STM32. We will be using *PuTTY* to do this.

Open PuTTY

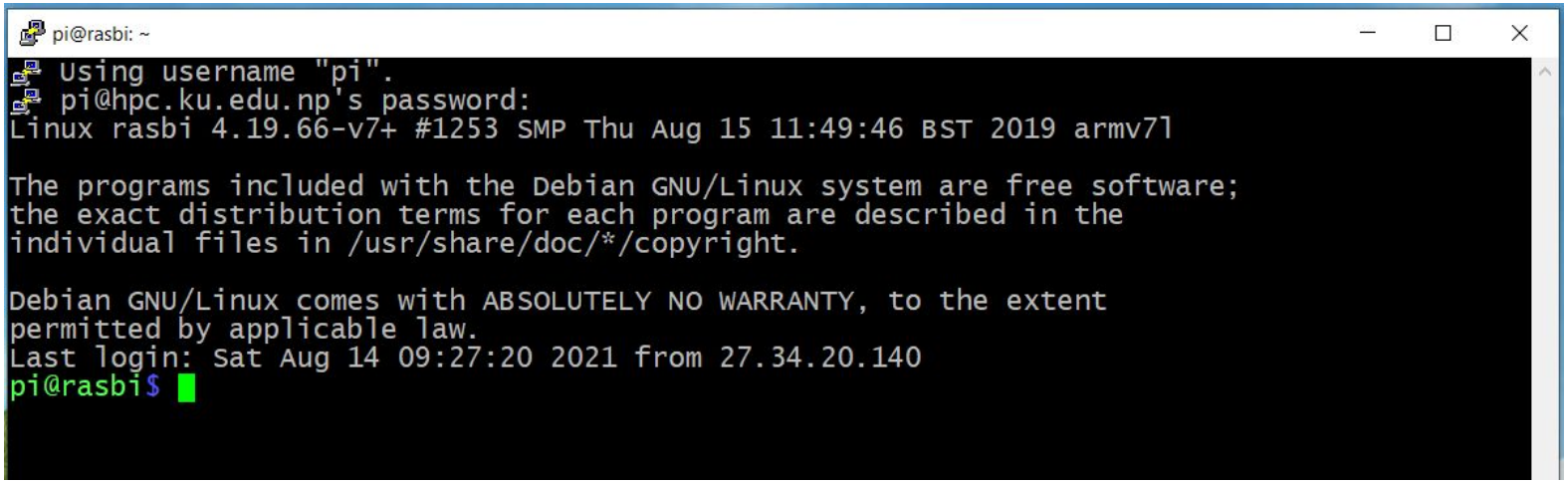

PuTTY Configuration
Category:
Session
Logging
Terminal
Keyboard
Bell
Features
Window
Appearance
Behaviour
Translation
Selection
Colours
Connection
Data
Proxy
SSH
Serial
Telnet
Rlogin
SUPDUP
Basic options for your PuTTY session
Specify the destination you want to connect to
Host Name (or IP address)
Port
22
Connection type:
SSH
Serial
Other:
Telnet
Load, save or delete a stored session
Saved Sessions
Default Settings
Addon-7
Addon-9
BalanceTriangle
Raspberry Pi
Raspberry Pi - FloatSat
floatsat
Load
Save
Delete
Close window on exit:
Always
Never
Only on clean exit
About
Help
Open
Cancel

Contact us for password !!

Enter Details



Login to Raspberry Pi



Use *ls* to view folders and files

pi@rasbi: ~

```
Using username "pi".
pi@hpc.ku.edu.np's password:
Linux rasbi 4.19.66-v7+ #1253 SMP Thu Aug 15 11:49:46 BST 2019 armv7l

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Sat Aug 14 09:27:20 2021 from 27.34.20.140
pi@rasbi$ ls
FloatSat  minicom.log  stlink.git
pi@rasbi$
```

Go inside *FloatSat* folder

pi@rasbi: ~/FloatSat

```
Using username "pi".
pi@hpc.ku.edu.np's password:
Linux rasbi 4.19.66-v7+ #1253 SMP Thu Aug 15 11:49:46 BST 2019 armv7l

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permitted by applicable law.
Last login: Sat Aug 14 09:27:20 2021 from 27.34.20.140
pi@rasbi$ ls
FloatSat  minicom.log  stlink.git
pi@rasbi$ cd FloatSat
pi@rasbi$ ls
main.hex  Makefile
pi@rasbi$
```

You are all done. Use *make flash* command to flash code in STM32.

pi@rasbi: ~/FloatSat

```
pi@rasbi$ make flash
sudo st-flash write main.hex 0x08000000
st-flash 1.7.0-92-g165776a
/etc/stlink/chips: No such file or directory
2021-08-14T10:10:52 INFO common.c: F4x5/F4x7: 192 KiB SRAM, 1024 KiB flash in at least 16 KiB pages.
file main.hex md5 checksum: 7a918d4dd4d23321f05f811b63682b1f, stlink checksum: 0x0098c370
2021-08-14T10:10:52 INFO common.c: Attempting to write 187609 (0x2dcd9) bytes to stm32 address: 134217728 (0x8000000)
2021-08-14T10:10:52 WARN common.c: unaligned len 0x2dcd9 -- padding with zero
EraseFlash - Sector:0x0 Size:0x4000 2021-08-14T10:10:52 INFO common.c: Flash page at addr: 0x08000000 erased
EraseFlash - Sector:0x1 Size:0x4000 2021-08-14T10:10:53 INFO common.c: Flash page at addr: 0x08004000 erased
EraseFlash - Sector:0x2 Size:0x4000 2021-08-14T10:10:53 INFO common.c: Flash page at addr: 0x08008000 erased
EraseFlash - Sector:0x3 Size:0x4000 2021-08-14T10:10:53 INFO common.c: Flash page at addr: 0x0800c000 erased
EraseFlash - Sector:0x4 Size:0x10000 2021-08-14T10:10:55 INFO common.c: Flash page at addr: 0x08010000 erased
EraseFlash - Sector:0x5 Size:0x20000 2021-08-14T10:10:57 INFO common.c: Flash page at addr: 0x08020000 erased
2021-08-14T10:10:57 INFO common.c: Finished erasing 6 pages of 131072 (0x20000) bytes
2021-08-14T10:10:57 INFO common.c: Starting Flash write for F2/F4/F7/L4
2021-08-14T10:10:57 INFO flash_loader.c: Successfully loaded flash loader in sram
2021-08-14T10:10:57 INFO flash_loader.c: Clear DFSR
2021-08-14T10:10:57 INFO flash_loader.c: Clear CFSR
2021-08-14T10:10:57 INFO flash_loader.c: Clear HFSR
2021-08-14T10:10:57 INFO common.c: enabling 32-bit flash writes
2021-08-14T10:10:57 INFO common.c: Go to Thumb mode
2021-08-14T10:11:02 INFO common.c: Starting verification of write complete
2021-08-14T10:11:04 INFO common.c: Flash written and verified! jolly good!
sudo st-flash reset
st-flash 1.7.0-92-g165776a
/etc/stlink/chips: No such file or directory
2021-08-14T10:11:04 INFO common.c: F4x5/F4x7: 192 KiB SRAM, 1024 KiB flash in at least 16 KiB pages.
pi@rasbi$
```

Thank you !!